MATH 215 - Problem Set 04
Random Variables

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1. Independent trials consisting of the flipping of an unfair coin having probability 0.2 of coming up heads are continually performed until either a head occurs or a total of 5 flips is made.

   (a) If we let X denote the number of times the coin is flipped, then what are the probabilities of random variable X taking on one of the values 1, 2, 3, 4, and 5.

   (b) Write down and draw the PMF and CDF of the random variable X.

2. Suppose that an antique jewelry dealer is interested in purchasing a gold necklace for which the probabilities are 0.22, 0.36, 0.28 and 0.14, respectively, that she will be able to sell it for a profit of 250 dollars, sell it for a profit of 100 dollars, break even, or sell it for a loss of 150 dollars. If X is the random variable that represents the profit/loss, write down its probability mass function $p(x) = P\{X = x\}$ and its cumulative distribution function $F(x) = P\{X \leq x\}$. Draw both of the functions.

3. Determine the constant c so that:
$$p(x) = c\left(\frac{2}{3}\right)^x, \quad x \in \mathbb{Z}^+$$
satisfies the conditions of being a probability mass function.

4. X is a discrete random variable represented with the following PMF.
$$p(x) = \begin{cases} \frac{1}{4}, & x = 1 \\ \frac{1}{2}, & x = 2 \\ \frac{1}{8}, & x = 3 \\ \frac{1}{8}, & x = 4 \\ 0, & \text{elsewhere} \end{cases}$$
Find the expected value of X.

5. Two balls are chosen randomly from an urn containing 8 white, 4 black, and 2 orange balls. Suppose that we win \$2 for each black ball selected and we lose \$1 for each white ball selected. Let X denote our winnings. What are the possible values of X, and what are the probabilities associated with each value? Write and draw the probability mass function of X.

6. Two fair dice are rolled. Let X equal the product of the 2 dice. Compute $P\{X = i\}$ for $i = 1, \ldots, 36$.

7. Suppose that a die is rolled twice. What are the possible values that the following random variables can take on:

   (a) the maximum value to appear in the two rolls?

   (b) the minimum value to appear in the two rolls?

   (c) the value of the first roll minus the value of the second roll?

8. If the die in the previous question assumed to be fair, calculate the probabilities associated with the random variables in parts (a) through (c).

9. Suppose that the CDF of X is given by:

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{2} & 0 \leq x < 1 \\ \frac{3}{5} & 1 \leq x < 2 \\ \frac{4}{5} & 2 \leq x < 3 \\ \frac{9}{10} & 3 \leq x < 3.5 \\ 1 & x \geq 3.5 \end{cases}$$

   (a) Calculate the probability mass function of X.
   (b) Calculate $E[X]$.
   (c) Calculate $Var[X]$.

10. Four buses carrying 148 students from the same school arrive at a football stadium. The buses carry, respectively, 40, 33, 25, and 50 students. One of the students is randomly selected. Let X denote the number of students that were on the bus carrying the randomly selected student. One of the 4 bus drivers is also randomly selected. Let Y denote the number of students on the driver's bus.

   (a) Which of $E[X]$ or $E[Y]$ do you think is larger? Why?
   (b) Compute $E[X]$ and $E[Y]$.

11. A lottery ticket pays off at least the cost of the ticket with probability $p = 0.4$. Now suppose that you buy 5 tickets.

   (a) Find the probability that you will win back at least the cost of exactly 3 tickets.
   (b) If X is the random variate that represents the number of tickets that pay off at least its cost, write down its probability mass function $p(x) = P\{X = x\}$ and its cumulative distribution function $F(x) = P\{X \leq x\}$. Draw both of the functions.
   (c) Calculate $E[X]$ and $Var[X]$.

12. Two coins are to be flipped. The first coin will land on heads with probability 0.6, the second with probability 0.7. Assume that the results of the flips are independent, and let X equal the total number of heads that result.

   (a) Find $P(X = 1)$.
   (b) Determine $E[X]$.

13. It is known that 30% of a large population of mice inoculated with a serum are protected from a certain disease. If 4 mice are inoculated find the probability that:

   (a) none contracts the disease.
   (b) fewer than 2 contracts the disease.
   (c) more than 3 contracts the disease.
   (d) Let X denote the number of mice which contract the disease. Find and draw the probability mass function.
   (e) Draw the cumulative distribution function for X?
   (f) Calculate $E[X]$ and $Var[X]$.

14. Two fair dice are thrown independently. Let X_1 and X_2 be the outcome of the first and the second die respectively. Let random variable X denote the absolute difference of the outcomes, $X = |X_1 - X_2|$.

    (a) Write down the probability mass function $p(x) = P\{X = x\}$.

    (b) Write down the cumulative distribution function $F(x) = P\{X \leq x\}$.

    (c) Draw the CDF you have found in (b).

    (d) Calculate $E[X]$ and $Var[X]$.

    (e) After the rolls, our friend has told us the absolute difference is an even number. Then, what is the probability that this absolute difference was less than 3?

15. The lifetime in hours of a certain kind of radio tube is a random variable having a probability density function given by

$$f(x) = \begin{cases} 0 & x \leq 100 \\ \frac{100}{x^2} & x > 100 \end{cases}$$

    What is the probability that exactly 2 of 5 such tubes in a radio set will have to be replaced within the first 150 hours of operation? Assume that the events E_i, $i = 1, 2, 3, 4, 5$, that the ith such tube will have to be replaced within this time are independent.

16. Assume the following probability density function.

$$f(x) = \begin{cases} (x+c)/2 & x \in [-c, c] \\ 0 & \text{elsewhere} \end{cases}$$

    with $c > 0$.

    (a) Find the cumulative distribution function F.

    (b) Find $F(-0.5)$, $F(0)$ and $F(0.5)$.

17. Assume the following probability density function.

$$f(x) = \begin{cases} cx^3 & x \in [0, \frac{c}{4}] \\ 0 & \text{elsewhere} \end{cases}$$

    (a) Find c and draw the probability density function.

    (b) Find the cumulative distribution function F and draw it.

    (c) Find $F\left(\frac{c}{8}\right) = P\left\{X \leq \frac{c}{8}\right\}$.

18. Find the expectation of random variables, the density of which are given as:

(a) $f(x) = \begin{cases} \frac{x}{8}, & 0 \leq x \leq 4 \\ 0, & \text{elsewhere} \end{cases}$

(b) $f(x) = \begin{cases} \frac{1}{100} e^{-\frac{x}{100}}, & x \geq 0 \\ 0, & x < 0 \end{cases}$

(c) $f(x) = \begin{cases} \frac{3}{4}\left(2x - x^2\right), & 0 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$

(d) $f(x) = \begin{cases} \frac{1-x}{2}, & x \in [-1, 1] \\ 0, & \text{elsewhere} \end{cases}$

(e) $f(x) = \begin{cases} 1 + x, & -1 < x \leq 0 \\ 1 - x, & 0 < x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$

19. Find the variance of the random variables, the density of which are given as:

(a) $f(x) = \begin{cases} \frac{1-x}{2}, & x \in [-1, 1] \\ 0, & \text{elsewhere} \end{cases}$

(b) $f(x) = \begin{cases} 1 + x, & -1 < x \leq 0 \\ 1 - x, & 0 < x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$

20. Let X denote the ratio of the sand in a cement mixture and be randomly distributed between 0 and 1. The probability density function of X is:

$$f(x) = \begin{cases} 20x^k (1 - x) & x \in [0, 1] \\ 0 & \text{elsewhere} \end{cases}$$

where k is a positive integer.

(a) Find k and draw the probability density function.

(b) Find the cumulative distribution function F and draw it.

(c) $P\{0.25 \leq X \leq 0.75\}$.

21. Let X be a random variable with probability density function:

$$f(x) = \begin{cases} c\left(1 - x^2\right) & -1 \leq x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

(a) What is the value of c?

(b) What is the cumulative distribution function of X?

(c) Determine $E[X]$.

(d) Determine $Var[X]$.

22. The probability density function of X, the lifetime of a certain type of electronic device (measured in hours), is given by:

$$f(x) = \begin{cases} \frac{10}{x^2} & x > 10 \\ 0 & \text{x} \leq 10 \end{cases}$$

   (a) Find $P\{X \geq 20\}$.
   (b) Find the CDF of X.
   (c) What is the probability that, of 6 such types of devices with independent distributions, at least 3 will function for at least 15 hours?

23. The density function of X is given by:

$$f(x) = \begin{cases} a + bx^2 & 0 \leq x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

and the expectation of X is $E[X] = \frac{3}{5}$. Find a and b.

24. The lifetime in hours of an electronic tube is a random variable having a probability density function given by:

$$f(x) = xe^{-x}, \quad x \geq 0$$

Compute the expected lifetime of such a tube. Then compute the variance of the lifetime.

25. For a drilling machine in a machine shop, let X denote the percentage of time out of 40 hour workweek that the lathe is actually in use. It has the probability density function:

$$f(x) = 4(1-x)^3, \quad 0 \leq x \leq 1$$

Calculate $E[X]$ and $Var[X]$.

26. Consider the density function

$$f(x) = \begin{cases} k\sqrt{x} & x \in (0,2) \\ 0 & \text{elsewhere} \end{cases}$$

   (a) Evaluate k.
   (b) Find the cumulative distribution function F and use it to evaluate $P\{0.3 < X < 0.6\}$.
   (c) Find $E[X]$.
   (d) Find $Var[X]$.

27. For a random variable X, if $E[X] = 1$ and $Var[X] = 5$, find

   (a) $E[2 + 5X]$.
   (b) $Var[3X + 4]$.

28. In a soccer league, a certain team wins and loses, respectively 65% and 5% of the games they play at home. When the team is playing away from home, it is equally likely to win, lose, or have a tie.

- The team gains 3 league points in case of a win, 1 leauge point in case of a tie, and 0 points in case of a loss.
- The team plays half of their games at home, while the other half is played away from home.

Let X denote the amount of league points gained by this team in a random league game.

(a) Draw the cumulative distribution function of X.

(b) Calculate $Var[X]$.

29. Let X be a continuous random variable with the following probability density function:

$$f(x) = \begin{cases} ax + b, & 0 \leq x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$

where a and b are unknown real values. It is known that $P\{X \leq 1\} = 2P\{X \geq 2\}$.

(a) Compute $P\{1 \leq X \leq 2\}$. Show all of your calculations.

(b) Compute $E[X]$. Show all of your calculations.

30. Let X be a discrete random variable with the following cumulative distribution function:

$$F(x) = \begin{cases} 0, & x < -1.5 \\ a, & -1.5 \leq x < 0 \\ b, & 0 \leq x < 2.5 \\ 1, & x \geq 2.5 \end{cases}$$

where $a, b \in R$ are unknown constants. We also know that $E[X] = \frac{3}{4}$ and $Var[X] = \frac{45}{16}$. Provide a complete formulation for the probability mass function of X.

31. Let X be the lifetime of a component measured in hours, and let its probability density function be:

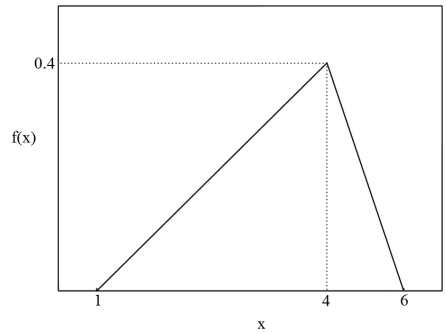
$$f(x) = 9xe^{-3x}, \quad x \geq 0$$

and zero elsewhere.

(a) What is the probability that the item works successfully for the first hour?

(b) Given that the item has worked successfully for the first hour, what is the probability that it does not fail within the next hour?

32. $f(x)$ is a probability density function, which follows a structure according to the following figure.



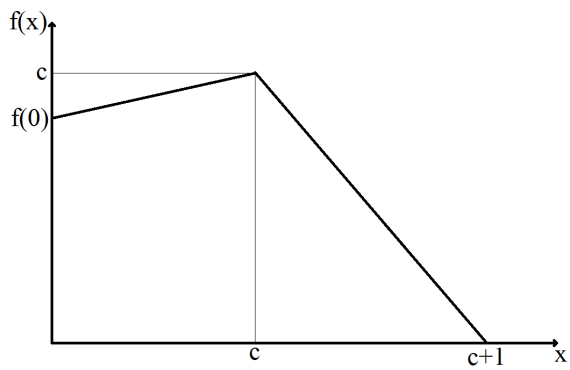
(a) Write down its probability density function in the closed form.

(b) Write down the closed form of the cumulative distribution function $F(x)$?

(c) Draw the the cumulative distribution function and indicate the values $F(1)$, $F(4)$ and $F(6)$.

(d) What is $P(3 < x < 5)$?

33. The following function f takes positive values only in $[0, c+1)$ and is zero elsewhere.



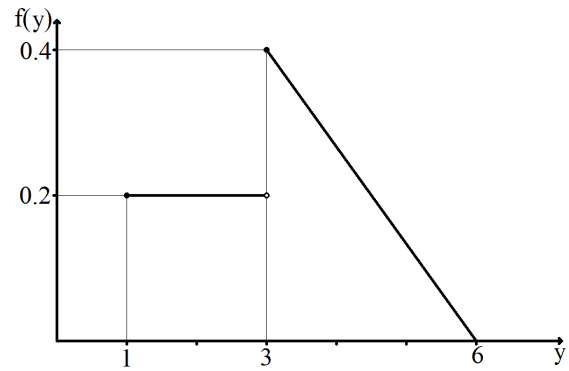
Suppose f forms a probability density function (PDF) for random variable X. Without calculating the CDF, without any integration, by using simple calculations:

(a) Formulate $f(0)$ in terms of c.

(b) If $f(0) = 0.7$, what is c?

(c) For the c value you have found in (b), find $P\left(\frac{c}{2} \leq X \leq 2c\right)$.

34. The random variable Y is distributed according to the following PDF.



(a) Write down the closed form of the PDF $f(y)$

(b) Write down the closed form of the CDF $F(y) = P(Y \leq y)$.

(c) Find $P(2 \leq Y \leq 5)$ by using the CDF you have found in (a).

35. A box contains 3 red and 7 black balls and we randomly choose 5 out of them. Let X_r and X_b denote the number of red and black balls in the chosen 5. We define the random variable $X = \max\{X_r, X_b\}$, namely the number of balls of the dominating color in the chosen five.

(a) Write down the probability mass function (PMF) $p(x) = P(X = x)$.

(b) Write down the cumulative distribution function (CDF) $F(x) = P(X \leq x)$.

(c) Find $E[X]$.

(d) Find $Var[X]$.

36. Let X be a continuous random variable with a linear density $f(x) = ax + b$ in the unit interval $[0, 1]$ and zero elsewhere. $E[X]$ is $\frac{5}{12}$. Find $Var[X]$.

37. X is a continuous random variable with the following density function:

$$f(x) = \begin{cases} x & 0 \leq x < 1 \\ x^{-3} & 1 \leq x < \infty \\ 0 & o/w \end{cases}$$

(a) Write down the closed form of the cumulative distribution function $F(x)$.

(b) Find $P(0.5 \leq X \leq 2)$.

(c) Find $E[X]$.

38. We toss a couple of fair dice. The game pays:

- \$1 if the upturned faces are equal,
- \$1 if there is at least one "six" among the upturned faces,
- \$1 if the sum of the upturned faces is equal to 8,
- and nothing for remaining outcomes.

According to above conditions, for some of the outcomes, it is possible to get more than \$1. What is the average payoff of this game.

39. Let X be a discrete random variable with the following cumulative distribution function:

$$F(x) = \begin{cases} 0, & x < -1 \\ \frac{1}{6}, & -1 \leq x < 0 \\ c, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$

where $c \in R$ is an unknown constant. If $E[X] = \frac{1}{3}$, what is $Var[X]$?

40. X is a random variable with the following probability mass function:

$$p(x) = \begin{cases} \frac{a-x}{c}, & x = 1, 2, 3, 4 \\ 0, & \text{elsewhere} \end{cases}$$

where $a, c \in R$ are unknown constant values. If $E[X] = 2$, what is $Var[X]$?

41. Suppose X is a random variable with the following probability mass function:

$$p(x) = \begin{cases} 0.1250, & x = 1 \\ a, & x = 2 \\ c, & x = 3 \\ 0.1875, & x = 4 \\ 0, & \text{elsewhere} \end{cases}$$

where $a, c \in R$ are unknown constant values. How many of the following values:

$$2.11, \quad 2.25, \quad 2.88, \quad 2.95, \quad 3.12, \quad 4.50$$

can $E[X]$ be equal to for suitable choices of a and c?

42. Suppose X is a random variable with the following cumulative distribution function:

$$F(x) = \begin{cases} 0, & x < -2 \\ 0.2, & -2 \leq x < -1 \\ a, & -1 \leq x < 0 \\ 0.6, & 0 \leq x < 1 \\ c, & 1 \leq x < 2 \\ 1, & x \geq 2 \end{cases}$$

where $a, c \in R$ are unknown constant values. What is $E[X]$?

43. Let X be a random variable with $E[X] = Var[X] = 5$. What is $E[X^2] + Var[3 - 2X]$?

44. Suppose X is a random variable with the following probability density function:

$$f(x) = \begin{cases} cx, & 0 \leq x \leq c \\ 0, & \text{elsewhere} \end{cases}$$

where $c \in R$ is an unknown constant. What is $E[X]$?

45. Suppose X is a random variable that follows the probability density function:

$$f(x) = \begin{cases} ax + b, & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

where $a, b \in R$ are unknown constant values. We know that $E[X] = \frac{2}{3}$. Then, what is $Var[X]$?

46. Consider the continuous random variable X with the following cumulative distribution function:

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{x^4}{16}, & 0 \leq x \leq 2 \\ 1, & x > 2 \end{cases}$$

What is $E[X]$?

47. Let X be a random variable with the following probability density function:

$$f(x) = \begin{cases} 6x(1-x), & 0 \leq x \leq 1 \\ 0, & \text{elsewehere} \end{cases}$$

What is $P\{|X - 0.5| > 0.25\}$?

48. Suppose X is a random variable with the following probability density function:

$$f(x) = \begin{cases} 0.4x, & 0 \leq x \leq 1 \\ 0.4, & 1 < x \leq 2 \\ \frac{3x^2}{140}, & 2 < x \leq 4 \\ 0, & \text{elsewhere} \end{cases}$$

What is $P\{X \leq 3\}$?

49. Suppose X is a random variable with the following probability density function:

$$f(x) = \begin{cases} \frac{x^3}{8}, & 0 \leq x < 2 \\ 1 - \frac{x}{4}, & 2 \leq x \leq 4 \\ 0, & \text{elsewhere} \end{cases}$$

What is $P\{1 < X \leq 3\}$?

50. Let X be a continuous random variable with the following probability density function:

$$f(x) = \begin{cases} 2c, & 0 \leq x \leq 2 \\ c, & 2 < x \leq 5 \\ 0, & \text{elsewhere} \end{cases}$$

where c is a real constant. Determine $P\{X > 1 | X \leq 4\}$.

51. Find the expected value of random variables X and Y represented with following PMFs.

(a) $p_X(x) = \begin{cases} \frac{2-x}{3}, & x = 0, 1 \\ 0, & \text{elsewhere} \end{cases}$

(b) $p_Y(y) = \begin{cases} \frac{y^2}{c}, & y = 1, 2, 3, 4 \\ 0, & \text{elsewhere} \end{cases}$

Solutions

1. (a) Let H_i denote the event of getting a head in the ith flip.

$$P(X=1) = P(H_1) = 0.2$$

$$P(X=2) = P(H_1^c \cap H_2) = P(H_1^c)\,P(H_2) = 0.8 \times 0.2 = 0.16$$

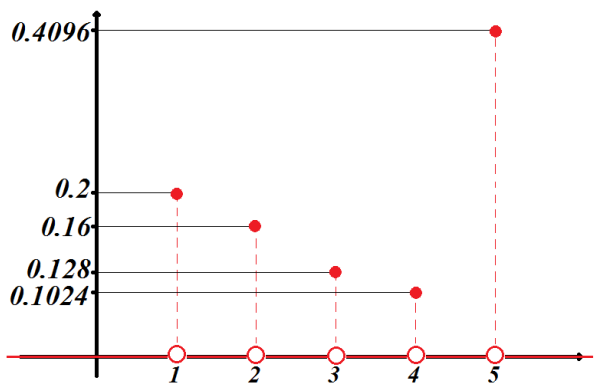
$$P(X=3) = P(H_1^c \cap H_2^c \cap H_3) = P(H_1^c)\,P(H_2^c)\,P(H_3) = (0.8)^2\,(0.2) = 0.128$$

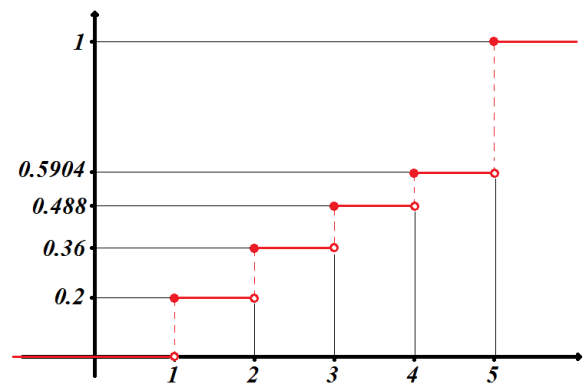
$$P(X=4) = P(H_1^c \cap H_2^c \cap H_3^c \cap H_4) = P(H_1^c)\,P(H_2^c)\,P(H_3^c)\,P(H_4) = (0.8)^3\,(0.2) = 0.1024$$

$$P(X=5) = P(H_1^c \cap H_2^c \cap H_3^c \cap H_4^c) = P(H_1^c)\,P(H_2^c)\,P(H_3^c)\,P(H_4^c) = (0.8)^4 = 0.4096$$

(b)

$$p(x) = \begin{cases} 0.2, & x=1 \\ 0.16, & x=2 \\ 0.128, & x=3 \\ 0.1024, & x=4 \\ 0.4096, & x=5 \\ 0, & \text{elsewhere} \end{cases} \qquad F(x) = \begin{cases} 0, & x<1 \\ 0.2, & 1 \le x < 2 \\ 0.36, & 2 \le x < 3 \\ 0.488, & 3 \le x < 4 \\ 0.5904, & 4 \le x < 5 \\ 1, & x \ge 5 \end{cases}$$





2. Drawing these functions is left to the reader as an exercise.

$$p(x) = \begin{cases} 0.14 & x=-150 \\ 0.28 & x=0 \\ 0.36 & x=100 \\ 0.22 & x=250 \\ 0 & \text{elsewhere} \end{cases} \qquad F(x) = \begin{cases} 0 & x<-150 \\ 0.14 & -150 \le x < 0 \\ 0.42 & 0 \le x < 100 \\ 0.78 & 100 \le x < 250 \\ 1 & x \ge 250 \end{cases}$$

3. Our golden rule: sum of all positive probability masses should be equal to one.

$$1 = \sum_{x=1}^{\infty} p(x) = \sum_{x=1}^{\infty} c\left(\frac{2}{3}\right)^x = c\sum_{x=1}^{\infty}\left(\frac{2}{3}\right)^x = c\left(\overbrace{\frac{2}{3} + \underbrace{\left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^3 + \ldots}_{\frac{2}{3}a}}^{a}\right)$$

$$a = \frac{2}{3} + \frac{2}{3}a \Rightarrow a = 2$$

$$ca = 2c = 1 \Rightarrow c = 0.5$$

4. $E[X] = \sum_{i=1}^{4} xp(x) = 1 \times \frac{1}{4} + 2 \times \frac{1}{2} + 3 \times \frac{1}{8} + 4 \times \frac{1}{8} = \frac{17}{8}$

5.

$$p(x) = P\{X = x\} = \begin{cases} 28/91, & x = -2 \\ 16/91, & x = -1 \\ 1/91, & x = 0 \\ 32/91, & x = 1 \\ 8/91, & x = 2 \\ 6/91, & x = 4 \\ 0, & \text{elsewhere} \end{cases}$$

6. Let us directly write down the probability mass function:

$$p(x) = \begin{cases} 1/36, & x = 1 \\ 2/36, & x = 2 \\ 2/36, & x = 3 \\ 3/36, & x = 4 \\ 2/36, & x = 5 \\ 4/36, & x = 6 \\ 2/36, & x = 8 \\ 1/36, & x = 9 \\ 2/36, & x = 10 \\ 4/36, & x = 12 \\ 2/36, & x = 15 \\ 1/36, & x = 16 \\ 2/36, & x = 18 \\ 2/36, & x = 20 \\ 2/36, & x = 24 \\ 1/36, & x = 25 \\ 2/36, & x = 30 \\ 1/36, & x = 36 \\ 0, & \text{elsewehere} \end{cases}$$

7. (a) $1, 2, 3, 4, 5, 6$
   (b) $1, 2, 3, 4, 5, 6$
   (c) $-5, -4, \ldots, 4, 5$

8. (a)

$$p(x) = \begin{cases} 1/36, & x = 1 \\ 3/36, & x = 2 \\ 5/36, & x = 3 \\ 7/36, & x = 4 \\ 9/36, & x = 5 \\ 11/36, & x = 6 \\ 0, & \text{elsewehere} \end{cases}$$

(b)

$$p(x) = \begin{cases} 11/36, & x = 1 \\ 9/36, & x = 2 \\ 7/36, & x = 3 \\ 5/36, & x = 4 \\ 3/36, & x = 5 \\ 1/36, & x = 6 \\ 0, & \text{elsewehere} \end{cases}$$

(c)

$$p(x) = \begin{cases} 1/36, & x = -5, 5 \\ 2/36, & x = -4, 4 \\ 3/36, & x = -3, 3 \\ 4/36, & x = -2, 2 \\ 5/36, & x = -1, 1 \\ 6/36, & x = 0 \\ 0, & \text{elsewehere} \end{cases}$$

9. (a) Calculating the jump sizes at respective points in the CDF help us to calculate probability mass values.

$$p(x) = P\{X = x\} = \begin{cases} 0.5, & x = 0 \\ 0.1, & x = 1 \\ 0.2, & x = 2 \\ 0.1, & x = 3 \\ 0.1, & x = 3.5 \\ 0, & \text{elsewhere} \end{cases}$$

(b) $E[X] = 0 \times 0.5 + 1 \times 0.1 + 2 \times 0.2 + 3 \times 0.1 + 3.5 \times 0.1 = 1.15$

(c) $E[X^2] = 0^2 \times 0.5 + 1^2 \times 0.1 + 2^2 \times 0.2 + 3^2 \times 0.1 + (3.5)^2 \times 0.1 = 3.025$

$$Var[X] = E\left[X^2\right] - E[X]^2 = 3.025 - (1.15)^2 = 1.7025$$

10. (a) $E[X]$ should be larger. While both random variables are taking the same possible values, X takes larger values with larger probabilities but Y takes all values with equally likely probabilities.

(b)

$$E[X] = 40 \times \frac{40}{148} + 33 \times \frac{33}{148} + 25 \times \frac{25}{148} + 50 \times \frac{50}{148} \approx 39.2838$$

$$E[Y] = 40 \times \frac{1}{4} + 33 \times \frac{1}{4} + 25 \times \frac{1}{4} + 50 \times \frac{1}{4} = 37$$

11. (a) Assuming independence between the results of tickets, we exactly need to win at three tickets with $(0.4)^3$ probability and exactly need to lose at two tickets with $(0.6)^2$ probability. However these three wins may come in any of the $\binom{5}{3}$ permutations. Thus the probability of interest is:

$$\binom{5}{3}(0.4)^3(0.6)^2 = 0.2304$$

(b) Drawing these functions is left to the reader as an exercise.

$$p(x) = \begin{cases} \binom{5}{0}(0.4)^0(0.6)^5 = 0.07776, & x = 0 \\ \binom{5}{1}(0.4)^1(0.6)^4 = 0.2592, & x = 1 \\ \binom{5}{2}(0.4)^2(0.6)^3 = 0.3456, & x = 2 \\ \binom{5}{3}(0.4)^3(0.6)^2 = 0.2304, & x = 3 \\ \binom{5}{4}(0.4)^4(0.6)^1 = 0.0768, & x = 4 \\ \binom{5}{5}(0.4)^5(0.6)^0 = 0.01024, & x = 5 \\ 0, & \text{elsewhere} \end{cases} \qquad F(x) = \begin{cases} 0, & x < 0 \\ 0.07776, & 0 \le x < 1 \\ 0.33696, & 1 \le x < 2 \\ 0.68256, & 2 \le x < 3 \\ 0.91296, & 3 \le x < 4 \\ 0.98976, & 4 \le x < 5 \\ 1, & x \ge 5 \end{cases}$$

(c)

$$E[X] = 0 \times 0.07776 + 1 \times 0.25920 + 2 \times 0.34560 + 3 \times 0.23040 + 4 \times 0.07680 + 5 \times 0.01024 = 2$$

$$E\left[X^2\right] = 0^2 \times 0.07776 + 1^2 \times 0.25920 + 2^2 \times 0.34560 + 3^2 \times 0.23040 + 4^2 \times 0.07680 + 5^2 \times 0.01024 = 5.2$$

$$Var[X] = E\left[X^2\right] - E[X]^2 = 5.2 - 2^2 = 1.2$$

12. Let H_i denote the event that the ith coin flip results with a head.

(a)

$$P(X = 1) = P(H_1 \cap H_2^c) + P(H_1^c \cap H_2) = P(H_1)P(H_2^c) + P(H_1^c)P(H_2) = 0.6 \times 0.3 + 0.4 \times 0.7 = 0.46$$

(b)

$$P(X = 0) = P(H_1^c \cap H_2^c) = P(H_1^c)P(H_2^c) = 0.4 \times 0.3 = 0.12$$

$$P(X = 2) = P(H_1 \cap H_2) = P(H_1)P(H_2) = 0.6 \times 0.7 = 0.42$$

$$E[X] = 0 \times 0.12 + 1 \times 0.46 + 2 \times 0.42 = 1.3$$

13. Assuming independence between mice (you can assume this if the mice population is sufficiently large), out of 4 mice, the probability that exactly x of them to contract the disease is:

$$\binom{4}{x}(0.7)^x(0.3)^{4-x}, \qquad x = 0, 1, 2, 3, 4$$

(a) $\binom{4}{0}(0.7)^0(0.3)^4 = 0.0081$

(b) $\binom{4}{0}(0.7)^0(0.3)^4 + \binom{4}{1}(0.7)^1(0.3)^3 = 0.0837$

(c) $\binom{4}{4}(0.7)^4(0.3)^0 = 0.2401$

(d) Drawing the probability mass function is left to the reader as an exercise. You can also answer part (a) and (c) by looking at the PMF.

$$p(x) = \begin{cases} 0.0081, & x = 0 \\ 0.0756, & x = 1 \\ 0.2646, & x = 2 \\ 0.4116, & x = 3 \\ 0.2401, & x = 4 \\ 0, & \text{elsewhere} \end{cases}$$

(e) Drawing the cumulative distribution function is left to the reader as an exercise. You can also answer part (b) by looking at the CDF.

$$F(x) = \begin{cases} 0, & x < 0 \\ 0.0081, & 0 \le x < 1 \\ 0.0837, & 1 \le x < 2 \\ 0.3483, & 2 \le x < 3 \\ 0.7599, & 3 \le x < 4 \\ 1, & x \ge 4 \end{cases}$$

(f) $E[X] = 2.8$ and $Var[X] = 0.84$. Calculation of those is left to the reader as an exercise.

14. (a)

$$p(x) = P\{X = x\} = \begin{cases} 6/36, & x = 0 \\ 10/36, & x = 1 \\ 8/36, & x = 2 \\ 6/36, & x = 3 \\ 4/36, & x = 4 \\ 2/36, & x = 5 \\ 0, & \text{elsewhere} \end{cases}$$

(b)

$$F(x) = P\{X \le x\} = \begin{cases} 0, & x < 0 \\ 6/36, & 0 \le x < 1 \\ 16/36, & 1 \le x < 2 \\ 24/36, & 2 \le x < 3 \\ 30/36, & 3 \le x < 4 \\ 34/36, & 4 \le x < 5 \\ 1, & x \ge 5 \end{cases}$$

(c) Drawing the cumulative distribution function is left to the reader as an exercise.

(d)

$$E[X] = 0 \times \frac{6}{36} + 1 \times \frac{10}{36} + 2 \times \frac{8}{36} + 3 \times \frac{6}{36} + 4 \times \frac{4}{36} + 5 \times \frac{2}{36} = \frac{35}{18}$$

$$E\left[X^2\right] = 0^2 \times \frac{6}{36} + 1^2 \times \frac{10}{36} + 2^2 \times \frac{8}{36} + 3^2 \times \frac{6}{36} + 4^2 \times \frac{4}{36} + 5^2 \times \frac{2}{36} = \frac{35}{6}$$

$$Var[X] = E\left[X^2\right] - E[X]^2 = \frac{35}{6} - \left(\frac{35}{18}\right)^2 = \frac{35}{6} - \frac{1225}{324} = \frac{665}{324}$$

(e)

$$P\{X < 3 | X \text{ is even}\} = \frac{P\{X < 3 \cap X \text{ is even}\}}{X \text{ is even}} = \frac{P\{X \in \{0,2\}\}}{P\{X \in \{0,2,4\}\}} = \frac{\frac{14}{36}}{\frac{18}{36}} = \frac{7}{9} \approx 0.7777$$

15. Let X denote the lifetime of an arbitrary radio. From the statement of the problem, we have:

$$P(E_i) = P(100 < X < 150) = \int_{100}^{150} f(x)\,dx = \int_{100}^{150} \frac{100}{x^2} dx = \int_{100}^{150} -\frac{100}{x} dx = \left[-\frac{100}{x}\right]_{100}^{150} = \left[-\frac{2}{3}\right] - [-1] = \frac{1}{3}$$

Thus, each radio tube is likely to be replaced within 150 hours with $\frac{1}{3}$ probability, independently. The probability that 2 of 5 such tubes in a radio set will have to be replaced within the first 150 hours of operation is:

$$\binom{5}{2}\left(\frac{1}{3}\right)^2\left(1 - \frac{1}{3}\right)^3 = 10 \times \frac{1}{9} \times \frac{8}{27} = \frac{80}{243}$$

16. (a) Since f is a probability density function, we must have $\int_{-\infty}^{\infty} f(x)\,dx = 1$. However, we only have non-negative PDF values between in the interval $[-c, c]$, implying that:

$$\int_{-c}^{c} \frac{1}{2}(x+c)\,dx = \left[\frac{1}{2}\left(\frac{x^2}{2}+cx\right)\right]_{-c}^{c} = \left[\frac{1}{2}\left(\frac{c^2}{2}+c^2\right)\right] - \left[\frac{1}{2}\left(\frac{c^2}{2}-c^2\right)\right] = \frac{3}{4}c^2 + \frac{1}{4}c^2 = c^2 = 1$$

which shows that $c = \pm 1$. As it is stated by the question, $c > 0$, thus, $c = 1$. The PDF is actually:

$$f(x) = \frac{x+1}{2}, \quad \text{for } x \in [-1, 1]$$

For $x \in [-1, 1]$

$$F(x) = P\{X \le x\} = \int_{-1}^{x} \frac{t+1}{2} dt = \left[\frac{t^2+2t}{4}\right]_{-1}^{x} = \left[\frac{x^2+2x}{4}\right] - \left[\frac{1-2}{4}\right] = \frac{x^2+2x+1}{4} = \frac{(x+1)^2}{4}$$

The closed form CDF is:

$$F(x) = \begin{cases} 0, & x < -1 \\ \frac{(x+1)^2}{4}, & -1 \le x \le 1 \\ 1, & x > 1 \end{cases}$$

(b)

$$F(-0.5) = \frac{(-0.5+1)^2}{4} = \frac{1}{16}$$

$$F(0) = \frac{(0+1)^2}{4} = \frac{1}{4}$$

$$F(0.5) = \frac{(0.5+1)^2}{4} = \frac{9}{16}$$

17. (a) Our golden rule: the total area under the PDF should be equal to one.

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{0}^{c/4} cx^3 dx = \left[\frac{cx^4}{4}\right]_{0}^{c/4} = \frac{c^5}{4^5} = 1 \quad \Rightarrow \quad c = 4$$

Thus, the PDF is:

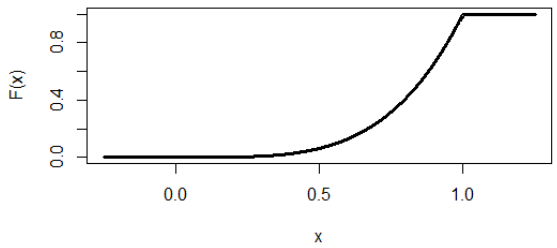
$$f(x) = \begin{cases} 4x^3 & x \in [0, 1] \\ 0 & \text{elsewhere} \end{cases}$$

(b) Assume $x \in [0, 1]$:

$$F(x) = P(X \le x) = \int_{0}^{x} 4t^3 dt = \left[t^4\right]_{0}^{x} = x^4$$

The closed form CDF is:

$$F(x) = \begin{cases} 0, & x < 0 \\ x^4, & 0 \le x \le 1 \\ 1, & x > 1 \end{cases}$$



(c) $F\left(\frac{c}{8}\right) = F\left(\frac{1}{2}\right) = P\left\{X \leq \frac{1}{2}\right\} = \frac{1}{16}$

18. For continuous random variables, the expected value is defined as:

$$E[X] = \int_{-\infty}^{\infty} xf(x)\,dx$$

(a)

$$E[X] = \int_0^4 \frac{x^2}{8}dx = \left[\frac{x^3}{24}\right]_0^4 = \frac{64}{24} = \frac{8}{3}$$

(b) The integral here requires integration by parts:

$$E[X] = \int_0^{\infty} \frac{x}{100}e^{-\frac{x}{100}}dx$$

Let $u = x$ and $dv = \frac{1}{100}e^{-\frac{x}{100}}$. We have:

$$du = dx, \qquad v = -e^{-\frac{x}{100}}$$

Then:

$$\int_0^{\infty} udv = [uv]_0^{\infty} - \int_0^{\infty} vdu = \left[-xe^{-\frac{x}{100}}\right]_0^{\infty} + \int_0^{\infty} e^{-\frac{x}{100}}dx$$

$\left[-xe^{-\frac{x}{100}}\right]_0^{\infty} = 0 - 0$, since at ∞, exponential function component converges to zero with a higher rate than x goes to infinity. The second component is:

$$\int_0^{\infty} e^{-\frac{x}{100}}dx = \left[-100e^{-\frac{x}{100}}\right]_0^{\infty} = 0 - (-100) = 100$$

Thus $E[X] = 100$

(c)

$$E[X] = \int_0^2 \frac{3}{4}\left(2x^2 - x^3\right)dx = \frac{3}{4}\left[\frac{2x^3}{3} - \frac{x^4}{4}\right]_0^2 = \frac{3}{4}\left[\frac{16}{3} - 4\right] - 0 = 1$$

(d)

$$E[X] = \int_{-1}^{1} \frac{x - x^2}{2}dx = \frac{1}{2}\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_{-1}^{1} = \frac{1}{2}\left[\frac{1}{6} - \frac{5}{6}\right] = -\frac{1}{3}$$

(e)

$$E[X] = \int_{-1}^{0} x + x^2 dx + \int_{0}^{1} x - x^2 dx = \left[\frac{x^2}{2} + \frac{x^3}{3}\right]_{-1}^{0} + \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_{0}^{1} = -\frac{1}{6} + \frac{1}{6} = 0$$

19. (a) $E[X]$ is already found as $-\frac{1}{3}$ in the previous question.

$$E\left[X^2\right] = \int_{-1}^{1} \frac{x^2 - x^3}{2} dx = \frac{1}{2}\left[\frac{x^3}{3} - \frac{x^4}{4}\right]_{-1}^{1} = \frac{1}{2}\left[\frac{1}{12} + \frac{7}{12}\right] = \frac{1}{3}$$

$$Var[X] = E\left[X^2\right] - E[X]^2 = \frac{1}{3} - \left(-\frac{1}{3}\right)^2 = \frac{2}{9}$$

(b) $E[X]$ is already found as zero in the previous question.

$$E\left[X^2\right] = \int_{-1}^{0} x^2 + x^3 dx + \int_{0}^{1} x^2 - x^3 dx = \left[\frac{x^3}{3} + \frac{x^4}{4}\right]_{-1}^{0} + \left[\frac{x^3}{3} - \frac{x^4}{4}\right]_{0}^{1} = \frac{1}{12} + \frac{1}{12} = \frac{1}{6}$$

$$Var[X] = E\left[X^2\right] - E[X]^2 = \frac{1}{6} - 0^2 = \frac{1}{6}$$

20. (a) The area under the PDF should be equal to 1.

$$\int_{0}^{1} 20\left(x^k - x^{k+1}\right) dx = 20\left[\frac{x^{k+1}}{k+1} - \frac{x^{k+2}}{k+2}\right]_{0}^{1} = 20\left(\frac{1}{k+1} - \frac{1}{k+2}\right) = 20\left(\frac{1}{k^2+3k+2}\right) = 1$$

This means:

$$\begin{aligned} \tfrac{1}{k^2+3k+2} = \tfrac{1}{20} \quad &\Rightarrow \quad k^2 + 3k + 2 = 20 \\ &\Rightarrow \quad k^2 + 3k - 18 = 0 \\ &\Rightarrow \quad k_{1,2} = \tfrac{-3 \pm \sqrt{(-3)^2 + 4 \times 18}}{2} \\ &\Rightarrow \quad k_{1,2} = \{-6, 3\} \end{aligned}$$

The true value for k is 3, as $k > 0$ is stated by the question. The PDF is:

$$f(x) = \begin{cases} 20x^3(1-x) & x \in [0,1] \\ 0 & \text{elsewhere} \end{cases}$$

Drawing the PDF is left to the reader as an exercise.

(b) For $x \in [0,1]$:

$$F(x) = \int_{0}^{x} 20\left(t^3 - t^4\right) dt = 20\left[\frac{t^4}{4} - \frac{t^5}{5}\right]_{0}^{x} = 5x^4 - 4x^5$$

The closed form CDF is:

$$F(x) = \begin{cases} 0, & x < 0 \\ 5x^4 - 4x^5, & 0 \le x \le 1 \\ 1, & x > 1 \end{cases}$$

(c) $P\{0.25 \le X \le 0.75\} = F(0.75) - F(0.25) \approx 0.6328 - 0.0156 \approx 0.6172$

21. (a) The area under the PDF should be equal to 1.

$$\int_{-1}^{1} c\left(1-x^2\right)dx = c\left[x-\frac{x^3}{3}\right]_{-1}^{1} = c\left(\frac{2}{3}+\frac{2}{3}\right) = \frac{4c}{3} = 1$$

Thus c should be $\frac{3}{4}$.

(b)

$$F(x) = \int_{-1}^{x} \frac{3}{4}\left(1-t^2\right)dt = \frac{3}{4}\left[t-\frac{t^3}{3}\right]_{-1}^{x} = \frac{-x^3+3x+2}{4}$$

The closed form CDF is:

$$F(x) = \begin{cases} 0, & x < -1 \\ \frac{-x^3+3x+2}{4}, & -1 \le x \le 1 \\ 1, & x > 1 \end{cases}$$

(c)

$$E[X] = \int_{-1}^{1} \frac{3}{4}\left(x-x^3\right)dx = \frac{3}{4}\left[\frac{x^2}{2}-\frac{x^4}{4}\right]_{-1}^{1} = \frac{3}{4}\left(\frac{1}{2}-\frac{1}{2}\right) = 0$$

(d)

$$E\left[X^2\right] = \int_{-1}^{1} \frac{3}{4}\left(x^2-x^4\right)dx = \frac{3}{4}\left[\frac{x^3}{3}-\frac{x^5}{5}\right]_{-1}^{1} = \frac{3}{4}\left(\frac{2}{15}+\frac{2}{15}\right) = 0.2$$

$$Var[X] = E\left[X^2\right] - E[X]^2 = 0.2 - 0^2 = 0.2$$

22. (a)

$$P\{X \ge 20\} = \int_{10}^{20} 10x^{-2}dx = \left[-10x^{-1}\right]_{10}^{20} = -\frac{1}{2} - (-1) = 0.5$$

(b) For $x \in [10, \infty)$:

$$F(x) = \int_{10}^{x} 10t^{-2}dt = \left[-10t^{-1}\right]_{10}^{x} = -\frac{10}{x} - (-1) = 1 - \frac{10}{x}$$

The closed form CDF is:

$$F(x) = \begin{cases} 0, & x < 10 \\ 1-\frac{10}{x}, & x \ge 10 \end{cases}$$

(c) Any device will function for at least 15 hours with probability:

$$P(X > 15) = 1 - F(15) = 1 - \left(1-\frac{10}{15}\right) = \frac{2}{3}$$

Assuming independence between devices, at least 3 out of 6 devices to function at least 15 hours is:

$$\binom{6}{3}\left(\frac{2}{3}\right)^3\left(\frac{1}{3}\right)^3 + \binom{6}{4}\left(\frac{2}{3}\right)^4\left(\frac{1}{3}\right)^2 + \binom{6}{5}\left(\frac{2}{3}\right)^5\left(\frac{1}{3}\right)^1 + \binom{6}{6}\left(\frac{2}{3}\right)^6\left(\frac{1}{3}\right)^0 = \approx 0.8999$$

23. The area under the PDF should be equal to 1.

$$\int_0^1 a + bx^2 dx = \left[ax + \frac{bx^3}{3}\right]_0^1 = a + \frac{b}{3} = 1$$

Also, the expected value of X is known:

$$E[X] = \int_0^1 ax + bx^3 dx = \left[\frac{ax^2}{2} + \frac{bx^4}{4}\right]_0^1 = \frac{a}{2} + \frac{b}{4} = \frac{3}{5}$$

As you can see, we have two linear equations with two unknowns (a and b), which are solved for:

$$a = 0.6, \qquad b = 1.2$$

24. We need to use integration by parts.

$$E[X] = \int_0^\infty x^2 e^{-x} dx$$

Say $u = x^2$ and $dv = e^{-x}dx$. Then $du = 2xdx$ and $v = -e^{-x}$.

$$E[X] = \int_0^\infty x^2 e^{-x} dx = \underbrace{\left[-x^2 e^{-x}\right]_0^\infty}_{0-0=0} + \int_0^\infty 2xe^{-x} dx$$

We can apply the same approach for the second integral. Say $u = 2x$ and $dv = e^{-x}dx$. Then $du = 2dx$ and $v = -e^{-x}$.

$$\int_0^\infty 2xe^{-x} dx = \underbrace{\left[-2xe^{-x}\right]_0^\infty}_{0-0=0} + \int_0^\infty 2e^{-x} dx = \left[-2e^{-x}\right]_0^\infty = 0 - (-2) = 2$$

Now let us calculate $E\left[X^2\right]$.

$$E\left[X^2\right] = \int_0^\infty x^3 e^{-x} dx$$

Using integration by parts again helps. Say $u = x^3$ and $dv = e^{-x}dx$. Then $du = 3x^2dx$ and $v = -e^{-x}$.

$$E\left[X^2\right] = \int_0^\infty x^3 e^{-x} dx = \underbrace{\left[-3x^2 e^{-x}\right]_0^\infty}_{0-0=0} + 3 \underbrace{\int_0^\infty x^2 e^{-x} dx}_{E[X]=2} = 6$$

Then the variance is:

$$Var[X] = E\left[X^2\right] - E[X]^2 = 6 - 2^2 = 2$$

25.

$$\begin{aligned} E[X] &= \int_0^1 4x(1-x)^3 dx \\ &= \int_0^1 4x(1-3x+3x^2-x^3)dx \\ &= \int_0^1 4x-12x^2+12x^3-4x^4 dx \\ &= \left[2x^2-4x^3+3x^4-\tfrac{4}{5}x^5\right]_0^1 \\ &= 0.2 \end{aligned}$$

On the average the lathe machine is in use for 8 hours.

$$\begin{aligned} E\left[X^2\right] &= \int_0^1 4x^2(1-x)^3 dx \\ &= \int_0^1 4x^2(1-3x+3x^2-x^3)dx \\ &= \int_0^1 4x^2-12x^3+12x^4-4x^5 dx \\ &= \left[\tfrac{4}{3}x^3-3x^4+\tfrac{12}{5}x^4-\tfrac{2}{3}x^6\right]_0^1 \\ &= \tfrac{1}{15} \end{aligned}$$

Then the variance is:

$$Var[X] = E\left[X^2\right] - E[X]^2 = \frac{1}{15} - \left(\frac{1}{5}\right)^2 = \frac{2}{75}$$

26. (a) The area under the PDF should be equal to 1.

$$\int_0^2 kx^{\frac{1}{2}}dx = \left[\frac{2k}{3}x^{\frac{3}{2}}\right]_0^2 = \frac{2k}{3}\sqrt{8} = 1 \quad \Rightarrow \quad k = \frac{3\sqrt{2}}{8}$$

(b) For $x \in (0,2)$:

$$F(x) = \int_0^x \frac{3\sqrt{2}}{8}t^{\frac{1}{2}}dx = \left[\frac{\sqrt{2}}{4}t^{\frac{3}{2}}\right]_0^x = \frac{\sqrt{2x^3}}{4}$$

The closed form CDF is:

$$F(x) = \begin{cases} 0, & x \le 0 \\ \frac{\sqrt{2x^3}}{4}, & 0 < x < 2 \\ 1, & x \ge 2 \end{cases}$$

The probability of interest is:

$$P\{0.3 < X < 0.6\} = F(0.6) - F(0.3) = \frac{\sqrt{0.432}}{4} - \frac{\sqrt{0.054}}{4} \approx 0.1062$$

(c)

$$E[X] = \int_0^2 \frac{3\sqrt{2}}{8}x^{\frac{3}{2}}dx = \frac{3\sqrt{2}}{8}\left[\frac{2}{5}\sqrt{x^5}\right]_0^2 = \frac{6}{5}$$

(d)

$$E\left[X^2\right] = \int_0^2 \frac{3\sqrt{2}}{8} x^{\frac{5}{2}} dx = \frac{3\sqrt{2}}{8}\left[\frac{2}{7}\sqrt{x^7}\right]_0^2 = \frac{12}{7}$$

Then the variance is:

$$Var\left[X\right] = E\left[X^2\right] - E\left[X\right]^2 = \frac{12}{7} - \left(\frac{6}{5}\right)^2 = \frac{300-252}{175} = \frac{48}{175}$$

27. (a) $E\left[2+5X\right] = 2 + 5E\left[X\right] = 7$

    (b) $Var\left[3X+4\right] = 3^2 Var\left[X\right] = 45$

28. (a) Let H denote the event that the team plays at home. We know that $P\left(H\right) = P\left(H^c\right) = 0.5$

$$P\left\{X=0\right\} = P\left(H\right)P\left\{X=0|H\right\} + P\left(H^c\right)P\left\{X=0|H^c\right\} = 0.5\left(0.05+\frac{1}{3}\right) = \frac{23}{120}$$

$$P\left\{X=1\right\} = P\left(H\right)P\left\{X=1|H\right\} + P\left(H^c\right)P\left\{X=1|H^c\right\} = 0.5\left(0.30+\frac{1}{3}\right) = \frac{38}{120}$$

$$P\left\{X=3\right\} = P\left(H\right)P\left\{X=3|H\right\} + P\left(H^c\right)P\left\{X=3|H^c\right\} = 0.5\left(0.65+\frac{1}{3}\right) = \frac{59}{120}$$

$$p_X\left(x\right) = \begin{cases} 23/120 & x=0 \\ 38/120 & x=1 \\ 59/120 & x=3 \\ 0 & \text{elsewhere} \end{cases} \qquad F_X\left(x\right) = \begin{cases} 0 & x<0 \\ 23/120 \approx 0.1917 & 0 \le x < 1 \\ 61/120 \approx 0.5083 & 1 \le x < 3 \\ 1 & x \ge 3 \end{cases}$$

The plot diagram for the CDF must be drawn accordingly with correct numbers and labels on axes.

(b) We can use the convenient formula of the variance.

$$E\left[X\right] = 0 \times \frac{23}{120} + 1 \times \frac{38}{120} + 3 \times \frac{59}{120} = \frac{43}{24}$$

$$E\left[X^2\right] = 0^2 \times \frac{23}{120} + 1^2 \times \frac{38}{120} + 3^2 \times \frac{59}{120} = \frac{569}{120}$$

$$Var\left[X\right] = E\left[X^2\right] - E\left[X\right]^2 = \frac{569}{120} - \left(\frac{43}{24}\right)^2 = \frac{4411}{2880} \approx 1.5316$$

29. (a) The area under the PDF must be equal to 1.

$$\int_0^3 ax + bdx = \frac{ax^2}{2} + bx\Big|_0^3 = \frac{9a}{2} + 3b = 1 \Rightarrow a = \frac{2-6b}{9}$$

$$P\{X \le 1\} = \int_0^1 ax + bdx = \frac{ax^2}{2} + bx\Big|_0^1 = \frac{a}{2} + b$$

$$P\{X \ge 2\} = \int_2^3 ax + bdx = \frac{ax^2}{2} + bx\Big|_2^3 = \frac{9a}{2} + 3b - 2a - 2b = \frac{5a}{2} + b$$

According to given information:

$$\frac{a}{2} + b = 2\left(\frac{5a}{2} + b\right) \Rightarrow a = \frac{-2b}{9}$$

Hence:

$$\frac{2-6b}{9} = \frac{-2b}{9} \Rightarrow b = \frac{1}{2} \Rightarrow a = -\frac{1}{9}$$

$P\{1 \le X \le 2\} = 1 - P\{X \le 1\} - P\{X \ge 2\} = 1 - \frac{4}{9} - \frac{2}{9} = \frac{1}{3}$

(b) We have determined that $f(x) = \frac{9-2x}{18}$ for $x \in [0, 3]$.

$$E[X] = \int_0^3 xf(x)\,dx = \int_0^3 \frac{9x - 2x^2}{18}dx = \frac{9x^2}{36} - \frac{2x^3}{54}\Big|_0^3 = \frac{5}{4} = 1.25$$

30. We have to transform to PMF:

$$P\{X = -1.5\} = P\{X \le -1.5\} - P\{X < -1.5\} = a - 0 = a$$

$$P\{X = 0\} = P\{X \le 0\} - P\{X < 0\} = b - a$$

$$P\{X = 2.5\} = P\{X \le 2.5\} - P\{X < 2.5\} = 1 - b$$

$$E[X] = -1.5 \times a + 0 \times (b-a) + 2.5 \times (1-b) = 2.5 - 1.5a - 2.5b = \frac{3}{4} \Rightarrow b = 0.7 - 0.6a$$

$$E\left[X^2\right] = 2.25 \times a + 0 \times (b-a) + 6.25 \times (1-b) = 6.25 + 2.25a - 6.25b = \frac{15}{8} + 6a$$

$$Var[X] = E\left[X^2\right] - E[X]^2 = \frac{15}{8} + 6a - \frac{9}{16} = \frac{96a + 21}{16} = \frac{45}{16} \Rightarrow a = 0.25,\ b = 0.55$$

Then the PMF is:

$$p(x) = \begin{cases} 0.25, & x = -1.5 \\ 0.30, & x = 0 \\ 0.45, & x = 2.5 \\ 0, & \text{elsewhere} \end{cases}$$

31. (a) Let us identify the CDF using integration by parts (say $u = 9t$ and $dv = e^{-3t}dt$):

$$F(x) = \int_0^x 9te^{-3t}dt = \left[-3te^{-3t}\right]_0^x + \int_0^x 3e^{-3t}dt = 1 - e^{-3x} - 3xe^{-3x}, \quad x \ge 0.$$

$$P\{X > 1\} = 1 - F(1) = e^{-3} + 3e^{-3} \approx 0.1991$$

(b) The question asks $P\{X > 2|X > 1\}$.

$$\begin{aligned} P\{X > 2|X > 1\} &= \frac{P\{X>2\cap X>1\}}{P\{X>1\}} = \frac{P\{X>2\}}{P\{X>1\}} \\ &= \frac{1-F(2)}{1-F(1)} = \frac{e^{-3\times 2}+3\times 2e^{-3\times 2}}{e^{-3}+3e^{-3}} \\ &= \frac{7}{4}e^{-3} \approx 0.0871 \end{aligned}$$

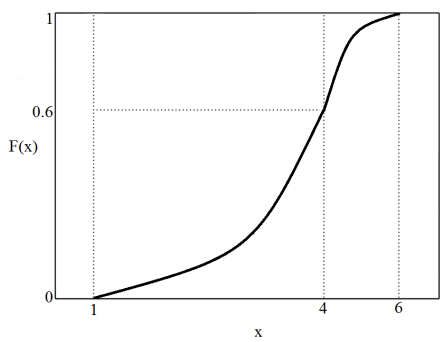
32. (a)

$$f(x) = \begin{cases} \frac{2}{15}(x-1), & 1 \le x \le 4 \\ \frac{6}{5} - \frac{1}{5}x, & 4 < x \le 6 \\ 0, & \text{elsewhere} \end{cases}$$

(b)

$$F(x) = \begin{cases} 0 & x \le 1 \\ \frac{1}{15}(x^2 - 2x + 1) & 1 < x \le 4 \\ \frac{3}{5} - \frac{1}{10}(x^2 - 12x + 32) & 4 < x \le 6 \\ 1 & x > 6 \end{cases}$$

(c) The CDF is shown in the following figure.



(d) $P(3 < x < 5) = F(5) - F(3) = \frac{9}{10} - \frac{4}{15} = \frac{19}{30}$

33. The PDF takes positive values only in $[0, c+1)$ and is zero elsewhere.

(a) The area below the function is:

$$\frac{cf(0) + c^2}{2} + \frac{c}{2} = 1 \quad \Rightarrow \quad f(0) = \frac{2 - c - c^2}{c}$$

(b) If $f(0) = 0.7$, what is c?

$$\begin{aligned} &\tfrac{2-c-c^2}{c} = 0.7 \quad \Rightarrow \quad c^2 + 1.7c - 2 = 0 \\ &c_{1,2} = \tfrac{-1.7\pm\sqrt{2.89+8}}{2} = \tfrac{-1.7\pm 3.3}{2} \\ &c_1 = 0.8, \quad c_2 = -2.5 \end{aligned}$$

Since the density values can never be negative, we are interested in a non-negative solution, thus $c = 0.8$.

(c) We calculate the area below the function in the interval $[0.4, 1.6]$. $f(0.4) = 0.75$, $f(1.6) = 0.16$.

$$\begin{aligned} P(0.4 \le X \le 1.6) &= \tfrac{0.75+0.8}{2} \times 0.4 + \tfrac{0.16+0.8}{2} \times 0.8 \\ &= 0.775 \times 0.4 + 0.48 \times 0.8 = 0.694 \end{aligned}$$

34. (a) The closed form of the PDF is:

$$f(y) = \begin{cases} 0.2 & 1 \leq y < 3 \\ -\frac{2}{15}y + 0.8 & 3 \leq y < 6 \\ 0 & o/w \end{cases}$$

(b)

$$F(y) = \begin{cases} 0 & y < 1 \\ \frac{y-1}{5} & 1 \leq y < 3 \\ \frac{-y^2+12y-21}{15} & 3 \leq y < 6 \\ 1 & y \geq 6 \end{cases}$$

(c) $F(5) - F(2) = \frac{14}{15} - \frac{1}{5} = \frac{11}{15}$

35. (a)

$$p(x) = \begin{cases} 6/12 & x = 3 \\ 5/12 & x = 4 \\ 1/12 & x = 5 \\ 0 & \text{elsewhere} \end{cases}$$

(b)

$$F(x) = \begin{cases} 0 & x < 3 \\ 6/12 & 3 \leq x < 4 \\ 11/12 & 4 \leq x < 5 \\ 1 & x \geq 5 \end{cases}$$

(c) $E[X] = 3 \times \frac{6}{12} + 4 \times \frac{5}{12} + 5 \times \frac{1}{12} = \frac{43}{12}$

(d)

$$E\left[X^2\right] = 9 \times \frac{6}{12} + 16 \times \frac{5}{12} + 25 \times \frac{1}{12} = \frac{159}{12}$$

$$Var[X] = E\left[X^2\right] - E[X]^2 = \frac{159}{12} - \left(\frac{43}{12}\right)^2 = \frac{59}{144}$$

36. Along with the given expectation, we also know that the area under the PDF should be zero. We can generate two equations for these two unknowns.

$$E[X] = \int_0^1 ax^2 + bxdx = \left[\frac{ax^3}{3} + \frac{bx^2}{2}\right]_0^1 = \frac{a}{3} + \frac{b}{2} = \frac{5}{12}$$

$$\int_0^1 ax + bdx = \left[\frac{ax^2}{2} + bx\right]_0^1 = \frac{a}{2} + b = 1 \Rightarrow a = -1, \quad b = \frac{3}{2}$$

$$E\left[X^2\right] = \int_0^1 -x^3 + \frac{3}{2}x^2dx = \left[-\frac{x^4}{4} + \frac{x^3}{2}\right]_0^1 = -\frac{1}{4} + \frac{1}{2} = \frac{1}{4}$$

$$Var[X] = E\left[X^2\right] - E[X]^2 = \frac{36}{144} - \frac{25}{144} = \frac{11}{144}$$

37. (a)

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{x^2}{2} & 0 \leq x < 1 \\ 1 - \frac{1}{2x^2} & 1 \leq x < \infty \end{cases}$$

(b) $P(0.5 \leq X \leq 2) = F(2) - F(0.5) = \frac{7}{8} - \frac{1}{8} = \frac{3}{4}$

(c) $E[X] = \int_0^1 x^2 dx + \int_1^{-\infty} x^{-2} dx = \left[\frac{x^3}{3}\right]_0^1 + \left[-\frac{1}{x}\right]_1^{-\infty} = \frac{1}{3} + 1 = \frac{4}{3}$

38. The outcomes of the sample space are related with the payment with random variable X, according to the following table.

X	1	2	3	4	5	6
1	\$1					\$1
2		\$1				\$2
3			\$1		\$1	\$1
4				\$2		\$1
5			\$1		\$1	\$1
6	\$1	\$2	\$1	\$1	\$1	\$2

$$E[X] = 0 \times P\{X = 0\} + 1 \times P\{X = 1\} + 2 \times P\{X = 2\} = \frac{0 \times 18 + 1 \times 14 + 2 \times 4}{36} \approx 0.6111$$

39. We have to transform to PMF:

$$P\{X = -1\} = P\{X \leq -1\} - P\{X < -1\} = \frac{1}{6} - 0 = \frac{1}{6}$$

$$P\{X = 0\} = P\{X \leq 0\} - P\{X < 0\} = c - \frac{1}{6}$$

$$P\{X = 1\} = P\{X \leq 1\} - P\{X < 1\} = 1 - c$$

$$E[X] = -1 \times \frac{1}{6} + 0 \times \left(c - \frac{1}{6}\right) + 1 \times (1 - c) = \frac{5}{6} - c = \frac{1}{3} \Rightarrow c = 0.5$$

$$E\left[X^2\right] = 1 \times \frac{1}{6} + 0 \times \frac{1}{3} + 1 \times \frac{1}{2} = \frac{2}{3}$$

$$Var[X] = E\left[X^2\right] - E[X]^2 \approx 0.5556$$

40.

$$\sum_{x=1}^{4} P\{X = x\} = \frac{4a - 10}{c} = 1$$

$$E[X] = 1 \times \frac{a-1}{c} + 2 \times \frac{a-2}{c} + 3 \times \frac{a-3}{c} + 4 \times \frac{a-4}{c} = \frac{10a - 30}{c} = 2$$

Two linear equations with two unknowns solves for $a = 5$ and $c = 10$.

$$E\left[X^2\right] = 1^2 \times \frac{4}{10} + 2^2 \times \frac{3}{10} + 3^2 \times \frac{2}{10} + 4^2 \times \frac{1}{10} = 5$$

$$Var[X] = E\left[X^2\right] - E[X]^2 = 5 - 2^2 = 1$$

41. By the properties of PMFs, we know that both a and c are non-negative and $a + c = 0.6875$. The minimum value of

$$E[X] = 1 \times 0.1250 + 2 \times 0.6875 + 3 \times 0 + 4 \times 0.1875 = 2.25$$

is attained when $a = 0.6875$ and $c = 0$. The maximum value of

$$E[X] = 1 \times 0.1250 + 2 \times 0 + 3 \times 0.6875 + 4 \times 0.1875 = 2.9375$$

is attained when $a = 0$ and $c = 0.6875$. That enables two possible values among candidates.

42. We have to transform to PMF:

$$P\{X = -2\} = P\{X \leq -2\} - P\{X < -2\} = 0.2 - 0 = 0.2$$

$$P\{X = -1\} = P\{X \leq -1\} - P\{X < -1\} = a - 0.2$$

$$P\{X = -0\} = P\{X \leq 0\} - P\{X < 0\} = 0.6 - a$$

$$P\{X = 1\} = P\{X \leq 1\} - P\{X < 1\} = c - 0.6$$

$$P\{X = 2\} = P\{X \leq 2\} - P\{X < 2\} = 1 - c$$

$$E[X] = -2 \times 0.2 - 1 \times (a - 0.2) + 0 \times (0.6 - a) + 1 \times (c - 0.6) + 2 \times (1 - c) = 1.2 - a - c$$

43. $E\left[X^2\right] = Var[X] + E[X]^2 = 5 + 25 = 30$ and $Var[3 - 2X] = (-2)^2 Var[X] = 20$. The answer is 50.

44. The area under the PDF is equal to 1. Therefore:

$$\int_0^c cx dx = \frac{c^3}{2} = 1 \Rightarrow c = 2^{\frac{1}{3}}$$

$$E[X] = \int_0^c cx^2 dx = \frac{c^4}{3} = \frac{2^{\frac{4}{3}}}{3} \approx 0.8399$$

45. The area under the PDF is equal to 1. Therefore we have:

$$\int_0^1 ax + b dx = \frac{a}{2} + b = 1$$

$$E[X] = \int_0^1 ax^2 + bx dx = \frac{a}{3} + \frac{b}{2} = \frac{2}{3}$$

two linear equations with two unknowns, which are solved for $a = 2$ and $b = 0$.

$$E\left[X^2\right] = \int_0^1 2x^3 dx = \frac{1}{2}$$

$$Var[X] = E\left[X^2\right] - E[X]^2 = \frac{1}{2} - \left(\frac{2}{3}\right)^2 \approx 0.0556$$

46. The PDF is $f(x) = F'(x) = \frac{x^3}{4}$ for $0 \leq x \leq 2$.

$$E[X] = \int_0^2 \frac{x^4}{4} dx = 1.6$$

47. $P\{|X - 0.5| > 0.25\} = P\{X < 0.25\} + P\{X > 0.75\} = \int_0^{0.25} 6x(1-x)\,dx + \int_{0.75}^{1} 6x(1-x)\,dx = 0.3125$

48. $P\{X \leq 3\} = 1 - P\{X > 3\} = 1 - \int_3^4 f(x)\,dx = 1 - \int_3^4 \frac{3x^2}{140} dx = 1 - \frac{64-27}{140} \approx 0.7357$

49. $P\{1 < X \leq 3\} = \int_1^3 f(x)\,dx = \int_1^2 \frac{x^3}{8} dx + \int_2^3 1 - \frac{x}{4} dx \approx 0.8438$

50.

$$P\{X > 1 | X \leq 4\} = \frac{P\{1 < X \leq 4\}}{P\{X \leq 4\}} = \frac{2c + 2c}{4c + 2c} \approx 0.6667$$

51. The expectations are simple to calculate.

    (a) $E[X] = 0 \times \frac{2}{3} + 1 \times \frac{1}{3} = \frac{1}{3}$

    (b) For p_Y to establish a PMF, the constant c must be equal to 30.

$$E[Y] = 1 \times \frac{1}{30} + 2 \times \frac{4}{30} + 3 \times \frac{9}{30} + 4 \times \frac{16}{30} = \frac{10}{3}$$